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Temporal Validity of Robotic Camera Observations: Host Update Age and Image–Joint Timestamp Audits

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Highlights

What are the main findings?

- A broader archive distinguished sparse near-30 Hz gaps from a near-15 Hz record that exceeded a fixed 50 ms age budget.
- Three message sessions showed that identical Header keys and different image–joint pairing rules imply different recorded temporal and joint-state relationships.

What are the implications of the main findings?

- Image–joint pairing should distinguish timestamp alignment from recorder-order availability.
- Exposure and consumer timing are needed to extend a host-log audit to full observation validity.

Abstract

Robotic sensing requires temporal guarantees beyond a regular update rate. We audited 17 original camera-stability files, 85 additional numeric records, and 24 image–joint archives from three complete historical sessions. The original 12 long streams covered 438.19 stream-hours and contained 326 intervals above 50 ms. The extension added 37 long records over 234.92 stream-hours, including a near-15 Hz record that spent 24.92% of recorded time above a 50 ms host update-age budget. Of 78,863 images, 78,860 uniquely matched CameraInfo Header stamps and frame IDs, yet median recording separations ranged from 9.65 to 191.66 ms across bags. For 78,825 eligible image–joint pairs, nearest-header matching gave median absolute differences of 2.43–2.59 ms; latest-recorded state selection gave signed differences of 15.85–187.34 ms. In 2771 fixed 1 Hz image samples, per-bag 95th percentiles of maximum recorded joint-position difference ranged from 0.0165 to 0.5049 degrees across 20 joints. Two additional head-camera contexts reproduced the temporal distinction. These findings separate update cadence, timestamp agreement, and recorder-order availability, supporting explicit timing budgets and pairing requirements. Exposure synchronization and physical spatial accuracy remain unmeasured.

Keywords: robotic sensing; camera timing; temporal validity; age of information; observation freshness; retrospective analysis; data provenance

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1. Introduction

Robotic perception depends on when an observation remains usable as well as on what it contains. Camera images may support localization, manipulation, or a learned policy while the robot and its surroundings continue to move. Exposure timing, transfer delay, application readout, and downstream consumption describe different points in that process. Work on visual–inertial sensing and heterogeneous camera synchronization

demonstrates why those distinctions matter for combining measurements [1–3]. A deployment log that records only application-level timestamps provides a narrower view, but it can still reveal whether image updates reach that application regularly.

Historical stability records are attractive for studying operational sensing because they can span much longer periods than a short benchmark run. Their scientific value depends on reconstructing the measurement contract. A column called a timestamp may represent read completion rather than exposure; a quantity labeled dropped frames may count threshold intervals; and a nominal camera configuration may be a script default rather than a preserved acquisition setting. Treating those fields as stronger measurements can turn a timing audit into an unsupported claim about sensor loss, synchronization, or spatial accuracy. Conversely, discarding all such records would forfeit useful evidence about the host-visible update process.

Even after the measurement layer is identified, summaries answer different questions. Mean frame rate describes throughput over a window, the 99th percentile (P99) describes a portion of the interval distribution, and a maximum exposes a single extreme. For an application that holds the latest observation between updates, the relevant temporal residence also depends on how long update age remains above a budget. Age-of-information theory provides established tools for that calculation [4–6]. Applying them to robotic deployment logs requires care with clock semantics, observation support, concurrent streams, and the independence of experimental units.

This study asks three questions. First, which temporal-validity statements survive source and byte-integrity checks of existing robotic camera records? Second, how do time weighting, periodic summaries, and concurrent-stream requirements characterize the same host-update process? Third, what do retained save positions and timestamped image–joint messages reveal about the limits of interpreting update recurrence as observation freshness?

An initial archive contains 17 stability CSV files from five robotic computers, including 12 long streams over 438.19 stream-hours. A roster-directed search adds 85 distinct numeric files, with 37 further long records over 234.92 stream-hours. Retained image names support a logger-side association check. Separately, 24 existing MCAP bags from three complete historical sessions provide image, CameraInfo, and joint-state timestamps and a defined sample of recorded joint positions. Two metadata-selected head-camera contexts from additional hosts provide a separate sensitivity check.

The contribution is a reproducible measurement audit that connects traceable curation to three practical distinctions: update cadence versus sparse interval tails, declared timestamp agreement versus recorder-order availability, and the recorded configuration selected by an image–joint pairing rule. The cohorts retain their own observation supports and clustering. No new age law or synchronization algorithm is proposed. The results bound what the existing records establish, while exposure accuracy, Cartesian error, and robot-task performance require measurements unavailable here.

1.1. Related Work and the Measurement Gap

Sensor timing has several distinct layers. Furgale et al. jointly estimated temporal and spatial calibration parameters for multisensor systems [7]. Qin and Shen estimated camera–inertial time offsets within visual–inertial optimization [8]. These approaches relate measurements that represent a common evolving physical state. They require information beyond the sequence of times at which a host application finishes reading an image. A regular host read cadence alone cannot validate their temporal calibration assumptions.

Hardware and optical timing references make stronger measurements possible. VersaVIS combines camera triggering, exposure compensation, and host clock translation in a

visual–inertial sensing platform [1]. The SenTiBoard separates measurement timing from data transfer and supports configurable sensor integration [2]. More recently, RocSync used a visually decoded LED clock to recover exposure windows and align heterogeneous camera recordings [3]. These studies directly address synchronization and, in the case of optical timing, observe the camera exposure process. The present analysis has no such reference and makes no comparative claim about synchronization accuracy. It examines host read recurrence and, in a separate message archive, the consequences of choosing different timestamp layers for image–joint pairing.

Curated datasets serve a related but different purpose. EuRoC and TUM VI provide calibrated visual–inertial recordings for odometry evaluation [9,10]. Their acquisition design makes them useful for studying estimation under known sensing arrangements. Pooling their camera timestamps with application-level read timestamps would mix measurement layers and would not enlarge the independent sample for our question. We therefore use these datasets as examples of stronger acquisition provenance, without presenting a public-dataset experiment that was not performed. Gis et al. examined timing and repeatability in a sensor-in-the-loop architecture [11]; their controlled replay setting likewise differs from the opportunistic historical records studied here.

The mathematical basis for time weighting is established age-of-information theory. Kaul et al. evaluated update systems using time-average age [4]; subsequent work treats delivery delay, update scheduling, and general age penalties [5,6]. We do not propose a new age metric or a new synchronization method. Our contribution is an empirical audit that connects those established functionals to the exact semantics and limitations of robotic camera logs, reports both event and time denominators, and preserves the clustering and missing-provenance boundaries of the available records.

2. Materials and Methods

2.1. Study Design and Record Selection

This retrospective observational study examines existing camera stability records from robotic computers. The analysis date was 19 October 2026. Five computers were selected because an earlier local inventory had identified stability-test directories on them; this is a targeted convenience sample, not a random sample of robots or cameras. The current device roster was used only to locate those computers. Acquisition consisted of read-only file listing, file-status checks, and downloading existing files. No camera was opened, no service was restarted, and no robot command or new physical experiment was performed for this study.

All CSV files in the five previously identified directories were retained. Files were transferred losslessly and compressed locally. SHA-256 digests were calculated over the uncompressed original bytes. File size and modification time were compared before and after transfer, and every subsequent analysis verified the archived digest. We excluded no file on the basis of its timing outcome. A byte-integrity rule, added after the initial complete-file screen, permits removal only of a contiguous trailing NUL suffix. The original is preserved, and any terminal numeric fragment would be retained only if all six fields were finite. The observed suffixes followed complete lines, so the derived prefixes preserve all complete numeric rows. Removed byte counts and prefix hashes are recorded separately. The principal cohort comprises records lasting at least one hour, a duration criterion fixed in the retrospective analysis plan before complete CSV outcomes were inspected. Shorter records are reported separately in Supplementary Table S1. The first row of each file initializes the logger and does not define an inter-update interval. Analysis therefore begins at the first timestamp and ends at the last, without extrapolating an unobserved final outage.

Concurrent files sharing a computer and file start date form a recording group. A file can span several calendar days; the group label does not imply a 24 h observation period. Host labels identify retrieval locations; they do not independently attest to the historical producer. Stream counts, frame counts, and elapsed stream-hours describe the amount of recorded information; they do not represent independent experimental replicates. Stream-hours sum the durations of concurrent files and consequently exceed unique wall-clock exposure. Files with different hashes need not correspond to different physical cameras. Camera serial numbers and historical device-to-file assignments were not available, so we report recorded streams rather than a verified number of distinct sensors.

2.2. Roster-Directed Archive Expansion

During the fifth revision, we revisited the live device roster and attempted read-only SSH/SFTP access to every endpoint marked as a robot and online. Newly online endpoints at the follow-up roster check were added to the access ledger. We recursively listed accessible home and data-storage trees, recording explicit exclusions for credentials, application caches, dependencies, and build/install artifacts. Connection failures, inaccessible directories, missing roots, and incomplete containers remain separate coverage outcomes. An online roster entry does not establish SSH access or data availability. No remote shell, robot API, sensor acquisition, or remote file mutation was used in this expansion. The private per-endpoint ledger documents coverage; the scientific sample remains a convenience sample of recoverable historical records.

All recoverable stability CSVs were retained. Two additional discovered files on a subsequently offline endpoint could not be retrieved and contribute no timing outcomes. We deduplicated original uncompressed bytes by SHA-256 against the initial cohort and other new files, then screened identical timestamp sequences at the saved microsecond precision. Copies remain provenance links rather than additional timing observations. We also screen overlapping timestamp sequences for truncated or re-exported copies, without inferring physical camera identity. Empty, short, malformed, and duplicate files remain in the inventory. Only finite, strictly increasing timestamp sequences permit whole-file age integration; the same trailing-NUL rule and one-hour duration stratum apply. Newly recovered source files were checked against the initial logger. The original 17-file cohort remains frozen, and the extension is reported separately as an exploratory robustness check. The first-hour sensitivity retains intervals ending within 3600 s and integrates their observed support. Full-record and first-hour summaries distinguish additional observation duration from a change in the diagnostic requirement. This extension does not create a controlled host comparison or a fleet-wide reliability estimate.

2.3. Measurement Semantics and Provenance Limits

Each CSV contains a consecutive successful-read counter, a host timestamp in seconds, a stored interval in milliseconds, a field named `instant_fps`, rolling jitter, and `drop_flag`. The recovered Python acquisition script is byte-identical across the five original hosts; current copies recovered on the seven extension hosts have the same digest. In its acquisition path, OpenCV `VideoCapture.read()` returns an image, after which `time.time()` is sampled. Failed reads are skipped after a short sleep and do not receive a CSV row. The same loop writes the CSV and can periodically save a JPEG. OpenCV's read operation combines grabbing and retrieving a frame [12]; its completion time is not an exposure timestamp.

The recovered defaults request 1280 $\times$ 720 pixels at 30 frames/s, perform 30 unrecorded warm-up reads, use a 30-frame window for displayed frame rate, and save an image every 5 s. These are *code defaults*, not verified historical settings. Historical com-

mand lines, sensor serial numbers, exposure, actual negotiated resolution, USB topology, runtime versions, and contemporaneous load were not retained in the CSV files. The present copy of the script is not a version-controlled attestation of the historical acquisition. We therefore test its consistency with the saved columns rather than claiming a complete reconstruction of the original setup.

The host timestamp is rounded to six decimal places. The stored interval is rounded separately to three decimal places in milliseconds. We reconstruct intervals from successive timestamps and check agreement with the stored column to a tolerance of 0.0016 ms, allowing for both roundings and floating-point representation. We also reconstruct the 30-timestamp sliding frame-rate estimate, the population standard deviation of the latest 30 intervals, and the threshold flag. The flag is generated when an interval exceeds 1.5 times the requested nominal period. At the 30 Hz code default this is 50 ms. The original cohort agrees with that flag rule. For newly recovered records, the requested rate remains unverified; the fixed 50 ms budget is analyzed separately from stored flags and observed cadence. An interval can span one or many nominal periods, and neither the successful-read counter nor this binary flag identifies how many exposures were lost. The field is consequently treated as an interval-exceedance indicator, not as a physical frame-loss label.

The clock source is also a measurement limitation. Python's wall clock can be adjusted, unlike its monotonic clock [13]. We screen for non-finite values, duplicate or backward timestamps, and discontinuities in the row counter. Positive clock adjustments cannot be distinguished from a long acquisition interval using these columns alone. All durations and age calculations below are conditional on the recorded wall clock representing elapsed time. Numerical agreement between two columns derived from the same clock is an integrity check, not independent clock validation.

2.4. Exploratory Audits of Retained Image Names and Multisensor Messages

Revision-stage archival analyses examine retained save positions, message timestamps, and recorded joint states. Their specifications and amendments are retained with the analysis code. They introduce no new robot operation and are not pooled with the endurance cohort as independent repetitions.

First, we listed retained JPEG names in the corresponding stability directories without reading image content. Each name encodes a successful-read counter and a local time of day. We considered all same-host CSV files and inferred a host-level integer-hour offset from filename–timestamp agreement. A candidate requires the same counter and a filename time between -2 microseconds and 50 ms after the CSV timestamp. The small negative allowance covers saved precision. Only uniquely assigned names enter the primary association analysis; ambiguous and unmatched names remain in the audit. We repeat the analysis with 1 and 5 ms upper bounds. The rule uses filename consistency rather than the interval outcome.

For a retained save at frame N, the expected write-associated interval ends at $N+1$, because the recovered loop saves after recording frame N. Its local comparator averages intervals ending at $N-2$, $N-1$, $N+2$, and $N+3$. Eligibility requires the entire $N-4$ through $N+5$ interval-endpoint neighborhood. We report the paired mean difference, threshold counts, and lag profile. Conditional ranges resample paired 15 min block sums 2000 times with seed 20261019. Retained filenames identify save-associated positions, not write duration, image validity, or the absence of unretained saves. The schedule is observational and time-dependent, so the comparison does not estimate the causal effect of writing an image.

Second, the structured message cohort extends the prior single-session audit using the two selected calibrator archives. It requires all four camera roles and both collection stages within a recovered session. Three selected sessions satisfy this design, contributing all 24 MCAP bags from two robotic computers: 10 and 20 September 2026 on one host, and 16 October 2026 on a second host. They comprise train and validation recordings for head, body, left-wrist, and right-wrist camera roles. These labels describe the historical collection stages; no model was fitted or evaluated here. Session manifests identify supervised physical collection. We analyze message metadata and a defined subset of recorded joint positions, without reusing the sessions' spatial-calibration accuracy results. Each session contributes all eight role/stage combinations. The two September sessions share a host; roles, stages, images, and messages are repeated observations, not independent robots. Results are stratified by session and bag and are reported separately from the endurance logs.

The MCAP schema distinguishes message recording time l, publish time p, and the embedded ROS Header stamp h. The format permits $p = l$ when publish time is unavailable [14]; equality would therefore not establish zero transport delay. Header stamp semantics also depend on the publisher. We retain the recorded differences $l - h$, $l - p$, and $p - h$ without identifying h as exposure time. All image, CameraInfo, joint-state, and joint-command message headers were retrieved using the uncompressed chunk indexes. Exact byte ranges and original offsets were archived locally, with range-archive digests and remote size/mtime checks. This verifies the retrieved ranges, not the unread image payloads or a full-file digest. A separate dynamic ROS 2 decoder checks the Header parser on an already retrieved leading file fragment. An independent CDR implementation decodes every retained complete ROS Header. CameraInfo-declared image dimensions are decoded from the same prefixes and checked against one complete CameraInfo message per bag; these dimensions do not validate unread image pixels.

For each image, timestamp-nearest pairing selects the joint-state Header stamp with minimum absolute difference, resolving ties toward the earlier stamp. We also select the most recently recorded joint message satisfying $l_j \leq l_I$. This second rule models a recorder-order latest-state lookup; it is not a trace of an actual controller. Comparisons require common image/joint recording support and image Header stamps inside joint Header support. A preceding-header comparator additionally requires $h_j \leq h_I$ and $l_j \leq l_I$. We distinguish a joint Header stamp later than the image stamp from a joint message recorded later than the image.

After the first partial decode revealed large image Header-to-recording differences, we added an explicitly exploratory cross-check. Images are matched to CameraInfo messages by exact Header stamp and frame ID, retaining only unique CameraInfo keys. Their recording-time difference $l_I - l_C$ cancels any shared Header-to-recorder offset. It measures message recording-time separation on the recorder clock, while exposure and downstream consumption remain unobserved. All matching failures, support exclusions, and per-bag distributions are retained. These context analyses are descriptive and have no frame-level significance tests.

As a separate extension, we froze a metadata-based selection from an interim snapshot of completed container indexes before retrieving additional message Headers. Eligible nonprimary bags required at least 60 s duration and at least 100 messages in each required class. They also required a complete stable index, uncompressed chunks, and modification at least one hour before selection. Paths explicitly labeled as dry-run, simulation, synthetic, mock, fixture, unit-test, or fake were excluded. The snapshot contained 21 eligible bags. Within each host/recorded-UTC-month stratum, we selected the minimum SHA-256 of endpoint identifier and path. The two selected head-camera records,

X1 and X2, come from two additional hosts. The frozen candidate list, exclusions, and deterministic rule are retained. We apply the same full-file Header, exact-key, and pairing checks. These incomplete-role contexts remain separate from M1–M3 and do not extend the joint-position sample. This archive-based selection is a sensitivity check, not random or independent fleet validation.

2.5. Recorded Configuration Discrepancy under Two Pairing Rules

To examine a consequence of the pairing choice in the recorded robot state, we select the first eligible image in each one-second recording-time bin of each bag, starting from its first eligible image. This fixed sampling rule was specified before reading the paired position values. For each selected image we retrieve the complete JointState messages selected by nearest-header and latest-recorded pairing. The archived JointState schema identifies named positions and declares radians for revolute joints. We additionally retrieve each bag's archived robot description and verify joint types there. The monitored set is restricted to the two head joints, seven joints per arm, and four trunk joints; only named joints declared revolute or continuous and present in both messages enter the comparison. Hand/finger values are excluded. Degree values follow the archived schema's declared radian convention; independent calibration of the reported joint positions is unavailable.

For monitored joint k, the discrepancy is $|q_{L,k} - q_{N,k}|$, using the principal angular difference for continuous joints. We report the maximum absolute discrepancy over the monitored joints for each sampled image, together with per-joint and per-bag medians and 95th percentiles (P95) in degrees. Full ROS 2 deserialization checks the retrieved messages. A sensitivity analysis omits the first five seconds of each bag's sampled images. This is a deterministic subsample of recorded configuration differences under two information requirements. It does not measure image–robot spatial registration error, controller error, or synchronization accuracy. Sampling at 1 Hz also cannot establish a maximum over all image times.

2.6. Host Update Age and Two Exceedance Denominators

Let $t_0, \ldots, t_m$ be successful-read completion timestamps within one stream, $d_i = t_i - t_{i-1} > 0$ its m intervals, and $D = t_m - t_0 = \sum_i d_i$ its observed duration. The observed update rate is $f = m/D$. We define host update age at an intermediate time t as

$$a_h(t) = t - t_i, \qquad t_i \le t < t_{i+1}. \tag{1}$$

This is the backward recurrence time of the host update process. It resets to zero at each successful read. Classical age of information instead uses the generation time of the freshest available observation [4,5]. Under causal delivery, ordered observations, and a valid common time base, generation precedes host read completion, so Equation (1) gives an optimistic age component. Without exposure timestamps or a measurement of subsequent consumer delay, it is not the full observation age. Duplicate images can also be delivered while a_h repeatedly resets; image freshness is not certified by this metric.

For a budget τ, the interval-exceedance fraction is

$$Q(\tau) = \frac{1}{m} \sum_{i=1}^{m} \mathbf{1}(d_i > \tau). \tag{2}$$

The corresponding time-exceedance fraction is the exact fraction of the observed interval over which host update age exceeds the budget:

$$F(\tau) = \frac{1}{D} \int_{t_0}^{t_m} \mathbf{1}\{a_h(t) > \tau\}\, dt = \frac{\sum_i (d_i - \tau)_+}{D}, \tag{3}$$

where $(x)_+ = \max(x, 0)$. Both are descriptive functionals of the same timestamps. Q answers how often a logged interval exceeds a threshold; F answers how much recorded time is spent beyond it. Their difference is not a classification accuracy comparison and has no universal direction. There is no additional physical experiment behind either statistic.

The time-average host update age is

$$\bar{a}_h = \frac{\sum_i d_i^2}{2D} = \frac{\bar{d}}{2}(1 + \mathrm{CV}_d^2), \qquad \bar{d} = D/m, \tag{4}$$

where CV_d uses the population variance of the finite recorded interval list. This identity follows from the area of the age triangles; it does not require independent or identically distributed intervals. The matched constant-gap reference has the same m and D, with age $\bar{d}/2$ and $F_{\mathrm{const}}(\tau) = (\bar{d} - \tau)_+/\bar{d}$. It is an analytic reference, not an observed competing system or a proposed synchronization algorithm. We further report the contribution of the largest $\lceil 0.001m \rceil$ intervals to $\sum_i d_i^2$ to quantify tail concentration.

The primary budget is 50 ms because it reproduces the existing logger's nominal 30 Hz flag. Sensitivity budgets are 33.33, 66.67, 100, 250, 500, and 1000 ms. These budgets are engineering scenarios, not independently established task safety limits. For a constant-speed held-observation scenario, a displacement allowance ϵ and speed $v > 0$ imply $\tau = \epsilon/v$; for example, 5 mm at 0.1 m/s corresponds to 50 ms. This relation only interprets a budget. It does not measure robot displacement, calibration error, or controller performance.

2.7. Concurrent-Stream Availability

For streams sharing a host and date, we restrict analysis to their common recorded interval $[L, U]$, where L is the latest start and U the earliest end. Each gap contributes an exceedance segment $[t_{i-1} + \tau, t_i)$ if $d_i > \tau$. We clip these segments to $[L, U]$ and calculate their union exactly. The resulting group fraction

$$F_{\cup}(\tau) = \frac{\left| \bigcup_j \{ t \in [L, U] : a_{h,j}(t) > \tau \} \right|}{U - L} \tag{5}$$

is the time during which at least one recorded stream exceeds the budget. This describes the availability of an all-stream update-age requirement. It does not estimate exposure synchrony, prove that the streams are physically distinct, or establish a shared fault. Pairwise intersections are retained as descriptive supplementary results. We do not apply cross-host timestamp alignment.

2.8. Temporal Variation, Robustness, and Statistical Interpretation

We integrate duration, age area, and exceedance time into consecutive 60 s bins, splitting every gap exactly at bin boundaries. This prevents a long gap from being assigned entirely to its endpoint bin. We display temporal variation and retain all bins, including the final partial bin. For the original cohort, the main sensitivity analyses omit intervals ending before 60 s and remove the ten largest intervals from each finite list. The first sensitivity retains any interval crossing the 60 s boundary; it is not a new warm-up measurement. The second changes the observation support and is explicitly a tail-influence calculation, not a repaired record or a plausible intervention. Timestamp-derived and stored intervals provide an additional numerical sensitivity check.

An exploratory periodic-summary comparison, added after the full-record results, groups intervals by their endpoint into 60 s bins and evaluates mean rate below 29 Hz, interval P99 above 50 ms, and any interval above 50 ms. Quantiles use NumPy's linear

interpolation. These interval-list summaries differ from the exact continuous-time bin integration above. Partial terminal lists are retained and then excluded in a sensitivity check. No fault label is attached to any criterion.

For a within-record variability diagnostic, we resample non-overlapping temporal blocks of 1, 5, or 15 min with replacement, keeping each block's additive duration, age area, and exceedance time together. Each resample contains the original number of blocks; the last partial block is retained. We use 2000 resamples per configuration, seed 20261019 plus the block length, and the 2.5th and 97.5th percentiles of the resampled ratio. Blocking follows the principle of retaining local dependence in time-series resampling [15]. These are *conditional block-resampling ranges*, not device-population confidence intervals. Long-term stationarity is not established, and block-length sensitivity is reported rather than selecting the narrowest range.

The primary results are exact descriptive quantities for the available finite records. There is no randomized treatment, verified independent camera sample, physical fault label, or task outcome. We therefore do not use frame-level significance tests, claim a device-level treatment effect, or interpret an absence of threshold events as a zero future failure probability. Computation uses Python, NumPy, pandas, SciPy, and Matplotlib; the exact environment, frozen analysis plan, input hashes, scripts, and machine-readable outputs accompany the manuscript.

3. Results

3.1. Record Integrity and Analyzable Support

The original archive contains 17 files with 2,282,614,383 original bytes. All transfers passed the size, modification-time, and SHA-256 checks. Byte-level inspection identified two files with pure trailing NUL suffixes of 3868 and 1522 bytes. Both suffixes began after a complete numeric line. Removing those 5390 non-observation bytes in derived copies preserved every complete numeric row; there was no incomplete numeric fragment to discard. The raw archives remain unchanged. This repair was documented as a retrospective amendment after the initial finite-value check stopped the analysis.

After that repair, all 47,310,322 numeric rows were finite and consecutive within their files, and every successive timestamp difference was positive. Reconstructed intervals agreed with the stored intervals within 0.001239 ms. Across all files, the 50 ms rule exactly reproduced every logged threshold flag. The maximum reconstruction differences were 0.001012 Hz for the rolling frame-rate field and 0.000946 ms for rolling jitter. These results support internal consistency with the recovered calculation path; they do not validate the acquisition clock against an independent reference.

Twelve long streams from three hosts form four groups by computer and file start date (Table 1). They contain 47,307,128 numeric rows and span 438.19 stream-hours. The remaining five short files, from three hosts, span 3.00–60.03 s and are described in Supplementary Table S1. Stream labels identify files within groups; C and D are different dates on the same host. Concurrent streams contribute separately to stream-hours, so 438.19 h is not a duration of independent, non-overlapping robot operation.

3.2. Original Cohort: Stable Central Timing with Sparse Events

In the original long-stream cohort, observed update rates ranged from 29.9743 to 29.9983 Hz, and interval P99 ranged from 33.558 to 33.935 ms. The largest interval was 96.867 ms in B1; no principal-cohort interval exceeded 100 ms. Thus, the records support predominantly regular host updates, rather than widespread or severe timing instability. Nevertheless, five streams contained a total of 326 intervals above 50 ms, while seven contained none (Table 2). A1 had the largest interval-exceedance fraction, 0.013101%, and the

Table 1. Original principal cohort. Groups were defined by computer and file start date; files can span several days. A and B are separate hosts; C and D share a third host. No camera serial-number mapping is implied by the stream count.

Group	Recorded date	Streams	Complete numeric rows	Stream-hours
A	25 May 2026	2	1 663 980	15.42
B	14 Jul 2026	2	1 980 585	18.35
C	23 Sep 2026	4	19 936 931	184.66
D	26 Sep 2026	4	23 725 632	219.76

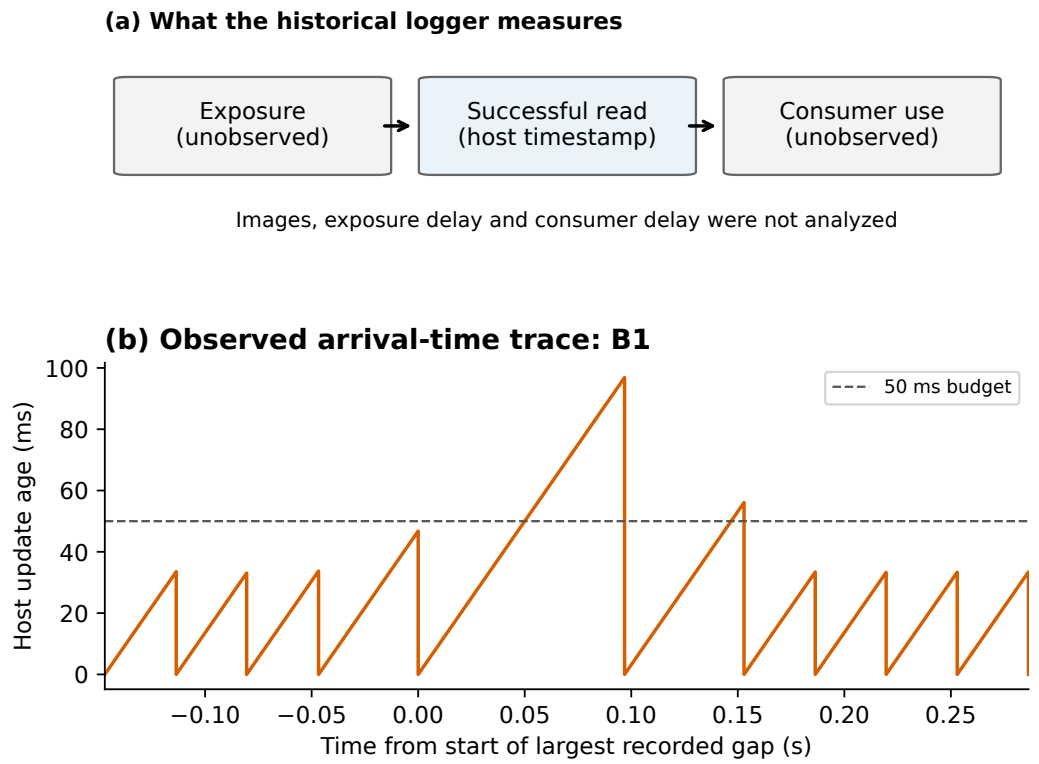


Figure 1. Measurement boundary and the host update-age functional. (a) The logger observes successful host read completion; exposure and consumer timing are absent. (b) The actual neighborhood of the largest original-cohort interval, in stream B1, illustrates the age reset and 50 ms budget. This is a trace-derived host-age curve, not an exposure-age measurement.

largest time-exceedance fraction, 0.005967%. Across the principal cohort, total host-age exceedance time at 50 ms was 4.869 s, or 0.000309% of stream-time.

The distinction between the two exceedance measures is visible even when both are small (Figure 2). In B2, four threshold intervals corresponded to 0.000404% of intervals, but only 0.000019% of recorded time. Counting an interval once does not encode how far it extends beyond the budget. Conversely, a percentile below the threshold does not exclude rarer events. The empirical tails are reported without fitting a heavy-tailed distribution.

Mean host update age ranged from 16.667692 to 16.683617 ms. Relative to the analytic constant-gap reference matched to each stream's count and duration, the largest age inflation was only 0.016%. The top 0.1% of intervals contributed 0.103955–0.146249% of the squared-gap sum. Thus, rare gaps were relevant to a strict 50 ms criterion but did not dominate average age in these records. The exact age identity and this small inflation are consistent with the overall regular timing, and prevent an isolated maximum from being interpreted as general degradation.

Table 2. Timing of all 12 original long streams. Rate uses the number of intervals divided by their duration. $n_{>50}$ counts intervals longer than 50 ms; $F(50)$ is the fraction of recorded time with host update age above 50 ms. These are different denominators.

Stream	Duration (h)	Rate (Hz)	P99 (ms)	Max (ms)	$n_{>50}$	$F(50)$ (%)
A1	7.71	29.9743	33.935	66.442	109	0.005967
A2	7.71	29.9857	33.926	36.155	0	0.000000
B1	9.18	29.9767	33.646	96.867	48	0.002213
B2	9.17	29.9855	33.633	53.403	4	0.000019
C1	46.22	29.9983	33.584	37.171	0	0.000000
C2	46.19	29.9983	33.579	37.423	0	0.000000
C3	46.17	29.9777	33.605	66.227	79	0.000713
C4	46.08	29.9854	33.598	38.179	0	0.000000
D1	54.96	29.9983	33.558	38.146	0	0.000000
D2	54.95	29.9777	33.576	65.520	86	0.000652
D3	54.93	29.9983	33.560	38.486	0	0.000000
D4	54.92	29.9854	33.569	37.838	0	0.000000

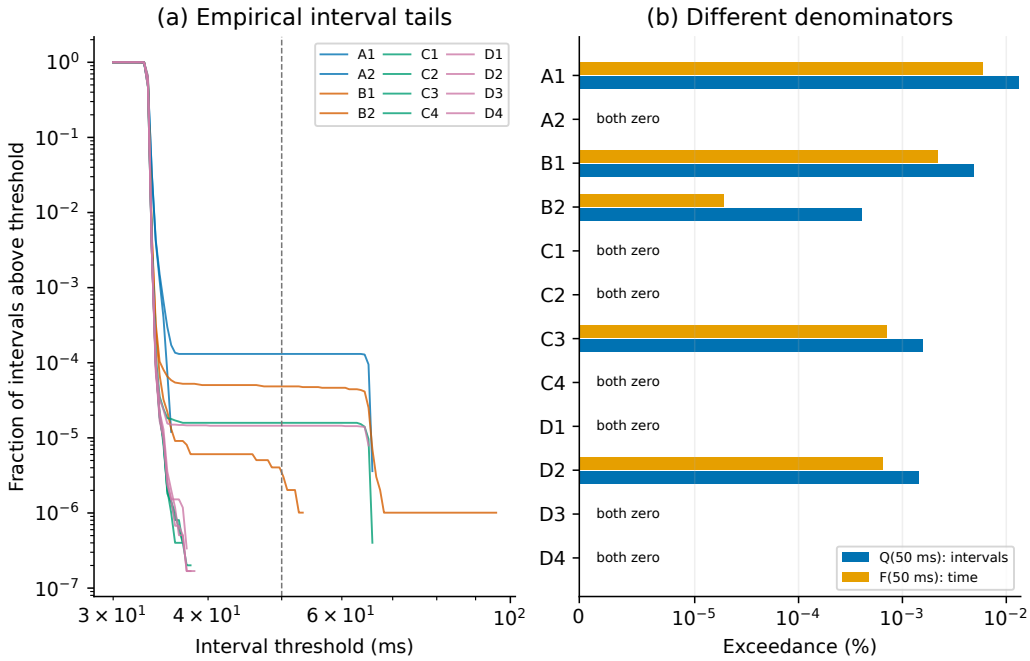


Figure 2. Interval tails and exceedance denominators for every original long stream. (a) Empirical survival evaluated on a logarithmic threshold grid; curves stop when the recorded exceedance count is zero. Color identifies the computer/file-start-date recording group. (b) Interval fraction $Q(50)$ and time fraction $F(50)$ on a symmetric logarithmic horizontal axis with a linear region near zero. Zero-event streams are retained.

3.3. Periodic Summaries Did Not Flag the Sparse Events

The exploratory periodic comparison tests the specified mean-rate and P99 criteria against an any-exceedance requirement, without independent physical fault labels.

The 326 threshold intervals occurred in 276 of 26,298 endpoint-assigned windows. Neither the mean-rate criterion nor the P99 criterion flagged any window, including the 276 containing an exceedance. Within those 276 windows, the lowest mean update rate was 29.8449 Hz and the highest P99 was 34.1445 ms. Excluding partial terminal windows retained 274 windows with threshold events and the same qualitative disagreement. Table 3 summarizes the comparison. A requirement to expose every 50 ms gap therefore needs a tail-sensitive criterion; neither a 29 Hz mean-rate gate nor a 50 ms P99 gate represents that requirement at this window length. This is criterion disagreement, not measured fault-detection sensitivity or a claim that one threshold is universally preferable.

Table 3. Exploratory offline summary comparison on 60 s endpoint-assigned interval lists. The event-containing subset is defined from the recorded intervals, without physical fault labels.

Criterion	All windows	Event-containing windows
Mean update rate < 29 Hz	0/26,298	0/276
Interval P99 > 50 ms	0/26,298	0/276
Any interval > 50 ms	276/26,298	276/276

Exact time integration over the record, rather than assigning entire gaps to endpoints, shows where exceedance residence occurred (Figure 3). These isolated nonzero periods coexist with many hours of zero exceedance. This temporal heterogeneity is retained in the resampling diagnostics rather than replaced with independent frame trials.

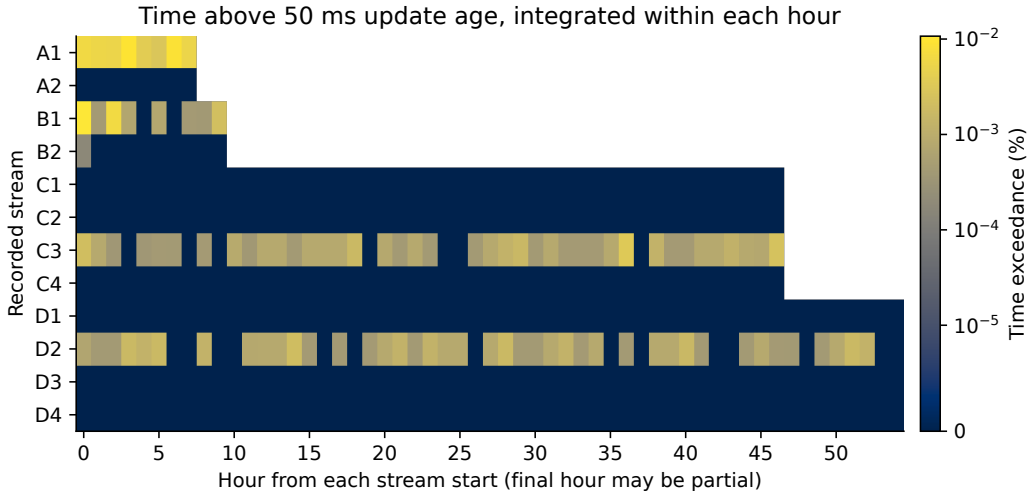


Figure 3. Hour-resolved time-exceedance fraction for all original long streams, obtained by summing exactly integrated 60 s age segments. The final hour can be partial. Zero values are retained; blank cells are beyond the observed support. The axes use each stream's elapsed time and do not imply cross-host clock synchronization.

3.4. Concurrent-Stream Availability and Sensitivity

The common recorded support of groups A–D was 7.70, 9.17, 46.08, and 54.91 h, respectively. Their union exceedance times at 50 ms were 1.656, 0.738, 1.155, and 1.290 s (Figure 4). No pairwise overlap of the above-budget host-age segments was observed within these common intervals. Consequently, the union duration equaled the sum of the

marginal durations in each group. In B, the second stream added a small non-overlapping contribution; in A, C, and D, all observed group exceedance came from one stream. These are statements about concurrent file timestamps, not evidence of independent physical camera failures or successful multisensor exposure synchronization.

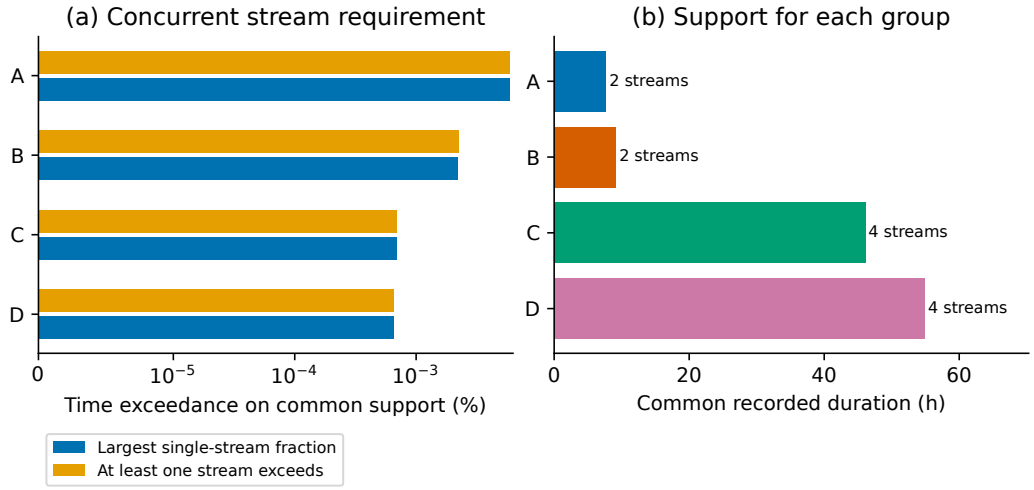


Figure 4. All-stream update-age availability on common same-host support. (a) The largest marginal time-exceedance fraction and the union fraction across all streams in each group. (b) The common recorded duration and number of streams. Group D shares a host with C. No cross-host comparison of timestamp phase is performed.

Sensitivity analysis at 66.67 ms retained only the 96.867 ms interval in B1; at 100 ms and higher budgets, all observed time-exceedance fractions were zero. The tighter 33.33 ms budget produced a different operating question because it lies near the nominal frame period. All tested budgets and stream-specific results are supplied in the analysis tables. Removing the first minute or the ten largest intervals did not change the conclusion of small mean-age inflation; the latter reduced tail contributions by construction and is not interpreted as a feasible correction. Supplementary Table S2 reports mean age, tail influence, and 15 min block-resampling ranges, while Supplementary Figures S1 and S2 show block-length and tail-removal diagnostics. For A1, the 15 min resampling range for $F(50)$ was 0.004789–0.007205%, around the observed 0.005967%. Zero-event records retained zero resampling ranges, which do not bound unobserved future events.

3.5. Few Long Intervals Immediately Followed Retained Saves

The filename audit on four original-cohort hosts found 315,411 JPEG names. Of these, 315,372 mapped uniquely to a CSV stream, 30 were ambiguous, and nine were unmatched. Every host's dominant filename offset was eight hours. The long-stream subset contained 315,350 retained saves, with median save spacing of 5.0003–5.0036 s across streams. This independently supports periodic saving in the historical records, while leaving the exact invocation and write duration unknown.

Among 315,338 eligible long-stream saves, three immediately following intervals exceeded 50 ms. The four local comparator positions contained eight exceedances among 1,261,352 intervals. Within-stream post-save mean differences were 0.47–10.31 microseconds. Stricter 1 and 5 ms filename gates retained the same three post-save exceedances and similarly small differences. These observations do not support a general account in which the 326 long intervals were produced by the retained image-save events. They also cannot rule out other logging effects or unretained saves. Supplementary Table S3 and Figure S3 give the full per-stream comparison and conditional ranges.

3.6. Archive Expansion Distinguished Cadence from Sparse Long Intervals

The roster-directed search recovered 90 additional stability-file paths on 7 further computers. Five paths contained only the identical CSV header: one unique header-only file and four byte-identical copies. The remaining 85 files contained 26,224,675 finite numeric rows over 246.80 stream-hours. Counters were consecutive, timestamps increased, and the fixed 50 ms rule reproduced the stored flags. No further trailing-NUL repairs or identical timestamp-sequence copies were found. The one-hour stratum retained 37 files on 4 hosts, with 24,956,026 rows over 234.92 stream-hours. All 48 shorter numeric records remain separate.

36 long extension records updated at 29.8925–29.9985 Hz and contained 1,312 intervals above 50 ms, totaling 29.000 s of above-budget host-age residence. Their largest interval was 665.723 ms, extending beyond the original cohort's observed tail. One 7.53 h record instead updated at 15.0161 Hz. Every one of its 407,320 intervals exceeded 50 ms, while host update age exceeded that budget for 24.9193% of recorded time (Table 4; Figure 5). When every interval exceeds τ, Equation (3) simplifies to $F(\tau) = 1 - \tau f$. Its large exceedance fraction therefore follows from cadence relative to the budget. The historical requested rate remains unknown; stored threshold flags do not count physical exposure loss.

Table 4. Exploratory long-record extension, separate from the original cohort. Ranges are across files within an observed-cadence stratum; a single value denotes one file. The fixed 50 ms budget is not a fault label.

Extension stratum	Long files	Stream-hours	Rate (Hz)	$Q(50)$ (%)	$F(50)$ (%)
Observed near 15 Hz	1	7.53	15.016	100.0000	24.91927
Observed near 30 Hz	36	227.38	29.893–29.998	0.0000–0.2869	0.00000–0.13032

In the near-30 Hz extension, 1,312 threshold intervals occupied 433 of 13,663 minute lists. The mean-rate criterion flagged none of those lists; P99 flagged 7. In the near-15 Hz record, all 453 lists contained threshold intervals and both criteria flagged every list. Periodic-summary disagreement therefore depends on the interval distribution and requirement. The lower-cadence record's first-hour time-exceedance fraction remained 24.9192%; at a 66.67 ms budget its full-record fraction fell to 0.0195%. Full budget-grid, short-file, startup, and conditional block-resampling results are retained.

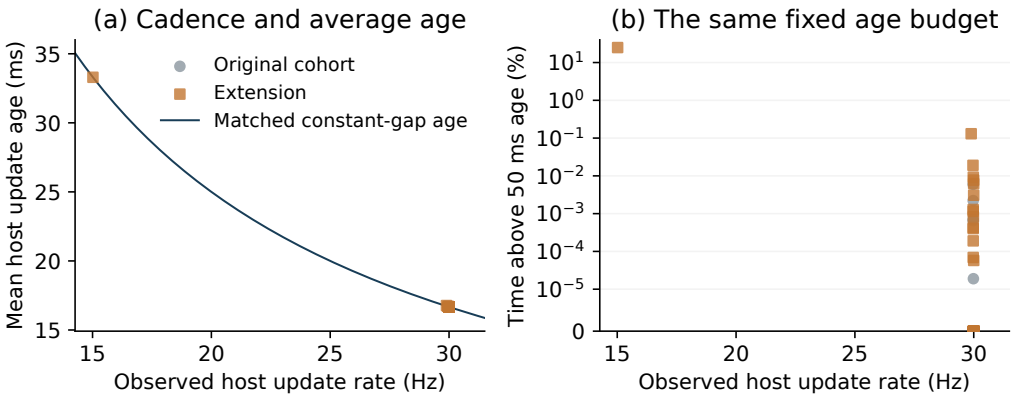


Figure 5. Original and exploratory long-record timing. (a) Mean host update age versus observed rate, with the analytic constant-gap reference. (b) Time above the same 50 ms age budget; a symmetric logarithmic axis retains zero values. Each marker is a file and overlapping markers are retained. Files are not verified independent cameras. The near-15 Hz record shows a cadence-related budget mismatch; near-30 Hz records contain sparse but heterogeneous interval tails.

3.7. Shared Message Timestamps Did Not Imply Coincident Recording

The complete message cohort comprises 24 bags from three historical sessions on two robotic computers, with 2762.05 bag-seconds, 78,863 image messages, and 276,159 joint-state messages. All 517,445 selected Headers matched the original indexes and topic counts. Independent CDR decoding agreed on every Header stamp and frame ID. These bag durations are separate from the camera stability stream-hours.

Sessions M1 and M2, dated 10 and 20 September 2026, share one host; M3, dated 16 October, belongs to the other host. Of 78,863 images, 78,860 matched unique CameraInfo messages with the same Header stamp and frame ID. Two M1 images and one M2 image lacked such a match and remain explicit exclusions from this comparison. All 83,742 CameraInfo records declared 640×480 pixels. Complete decoding of one deterministic sample per bag confirmed those fields; image pixels were not validated.

Per-bag median image-minus-CameraInfo recording differences were 9.65–11.56 ms in M1, 10.24–12.43 ms in M2, and 185.18–191.66 ms in M3 (Table 5; Figure 6a). Exact Header agreement therefore coexisted with recording-time separation in all three sessions, but its magnitude was session-dependent. The within-key comparison cancels a shared Header offset; it is not an exposure-offset measurement. Supplementary Table S4 separates recording, publish, and Header times and retains all per-bag diagnostics.

Table 5. Historical message cohorts. M1 and M2 share a host; M3 is from a second host. Every session includes train and validation collection stages for four camera roles. Shared keys count uniquely matched image/CameraInfo Header-stamp and frame-ID pairs. Difference columns give ranges of the eight per-bag medians: Δ_I is image-minus-CameraInfo recording time, $|\Delta_N|$ is absolute nearest-joint Header mismatch, and Δ_L is signed latest-recorded joint-minus-image Header mismatch. These are descriptive ranges, not confidence intervals.

Session	Bags	Images	Shared keys	Δ_I (ms)	$\lvert\Delta_N\rvert$ (ms)	Δ_L (ms)
M1	8	27,712	27,710	9.65–11.56	2.47–2.53	15.85–42.00
M2	8	27,789	27,788	10.24–12.43	2.43–2.53	16.89–42.78
M3	8	23,362	23,362	185.18–191.66	2.47–2.59	180.83–187.34

3.8. Image–Joint Pairing Changed with the Temporal Requirement

Common-support and preceding-state checks retained 78,825 images and excluded 38 boundary cases. Nearest-header joint pairing produced per-bag median absolute differences of 2.43–2.59 ms. Selecting the latest joint message already recorded at image recording time instead produced median signed differences of 15.85–42.78 ms in M1/M2 and 180.83–187.34 ms in M3 (Figure 6b). These quantities represent different information requirements, rather than competing estimates of physical alignment accuracy.

Across the retained pairs, 39,276 (49.83%) nearest joint Headers were later than the image Header. Nevertheless, none of those joint messages was recorded after the image. Every latest-recorded pairing also selected a joint Header later than the image Header. Recording order thus distinguishes a later declared timestamp from a message unavailable to the recorder at that instant. An independent pandas as-of implementation reproduced all 78,825 nearest and latest pairings with zero timestamp mismatches. Excluding the first 5 s retained the cross-session pattern; full preceding-header and startup results remain in the supplementary data.

Two metadata-selected head-camera cases on additional hosts retained 73,485 Headers and 11,397 eligible image–joint comparisons. Nearest-header medians were 2.54 and 2.56 ms; latest-recorded medians were 53.90 and 175.97 ms. Their exact-key recording separations and independent checks also supported the temporal distinction (Supplementary Table S8).

3.9. Pairing Choice Also Changed the Recorded Joint Configuration

The fixed 1 Hz rule selected 2771 eligible image times: 927 in M1, 930 in M2, and 914 in M3. Complete JointState decoding and the archived robot descriptions supported the same 20 monitored revolute joints in every bag. For each sampled image, we calculated the largest absolute position difference over those joints between the two selected messages. Per-bag P95 values ranged from 0.0193–0.1235° in M1, 0.0165–0.1338° in M2, and 0.1487–0.5049° in M3 (Figure 6c). The largest discrepancy among the sampled images was 0.9146°, in M3. Independent CDR parsing reproduced the position arrays and sampled differences (Supplementary Information).

The observed configuration differences depend on motion and timing, without identifying a causal host, publisher, or software effect. The 1 Hz sample does not bound every image-time discrepancy. Per-joint distributions, sample counts, and startup sensitivity remain in the supplementary data; Cartesian error and task performance were not measured.

4. Discussion

4.1. A Timing Budget Has to Be Interpreted against the Observed Cadence

The original cohort and its extension show why a sensing record needs an explicit temporal contract. Near-30 Hz updates were predominantly regular, with small mean-age inflation and sparse 50 ms intervals. The near-15 Hz extension record violated that budget throughout its interval list while spending approximately one quarter of recorded time above it. Those outcomes describe different relations between cadence and requirement. A binary threshold flag cannot, by itself, distinguish a lower sustained update rate from an isolated long interval or count lost physical exposures.

The periodic-summary contrast is similarly conditional. The original cohort had no specified periodic-summary flags, while P99 flagged a small subset of event-containing lists in the near-30 Hz extension. Both criteria flagged every list of the lower-cadence long record. This is consistent with the different requirements encoded by each statistic. A P99 criterion permits a smaller fraction of intervals beyond its threshold; a mean-rate criterion permits local variation within its window. An any-exceedance requirement addresses another question. Without physical fault labels or a task-defined budget, their disagreement is not a fault-detection accuracy result.

The established age framework makes the distinction quantitative. Equation (4) separates the effect of mean cadence from interval variability, while Equation (3) measures actual residence above a budget. Neither identity requires independent frame intervals. In the near-15 Hz record, the 50 ms budget lies below every observed gap, so residence follows directly from rate. In the original near-30 Hz records, sparse tails matter to a strict budget without dominating average age. This application of existing age-of-information theory [4,5] supports a measurement-aware audit rather than a new age law or scheduling method.

4.2. Shared Headers and Current-State Queries Describe Different Relationships

The message archive connects timestamp agreement to availability. Shared image and CameraInfo Header keys coexisted with session-dependent recording separation, even after cancelling their common Header offset. Timestamp agreement therefore did not establish coincident recorder availability. It does not localize the separation to encoding, buffering, transport, or scheduling. The common CameraInfo-declared dimensions also do not establish identical hardware, publisher configuration, or unread image content.

Image–joint pairing then poses a choice of information requirement. Nearest-header lookup targets numerical alignment with an image's declared timestamp; latest-recorded

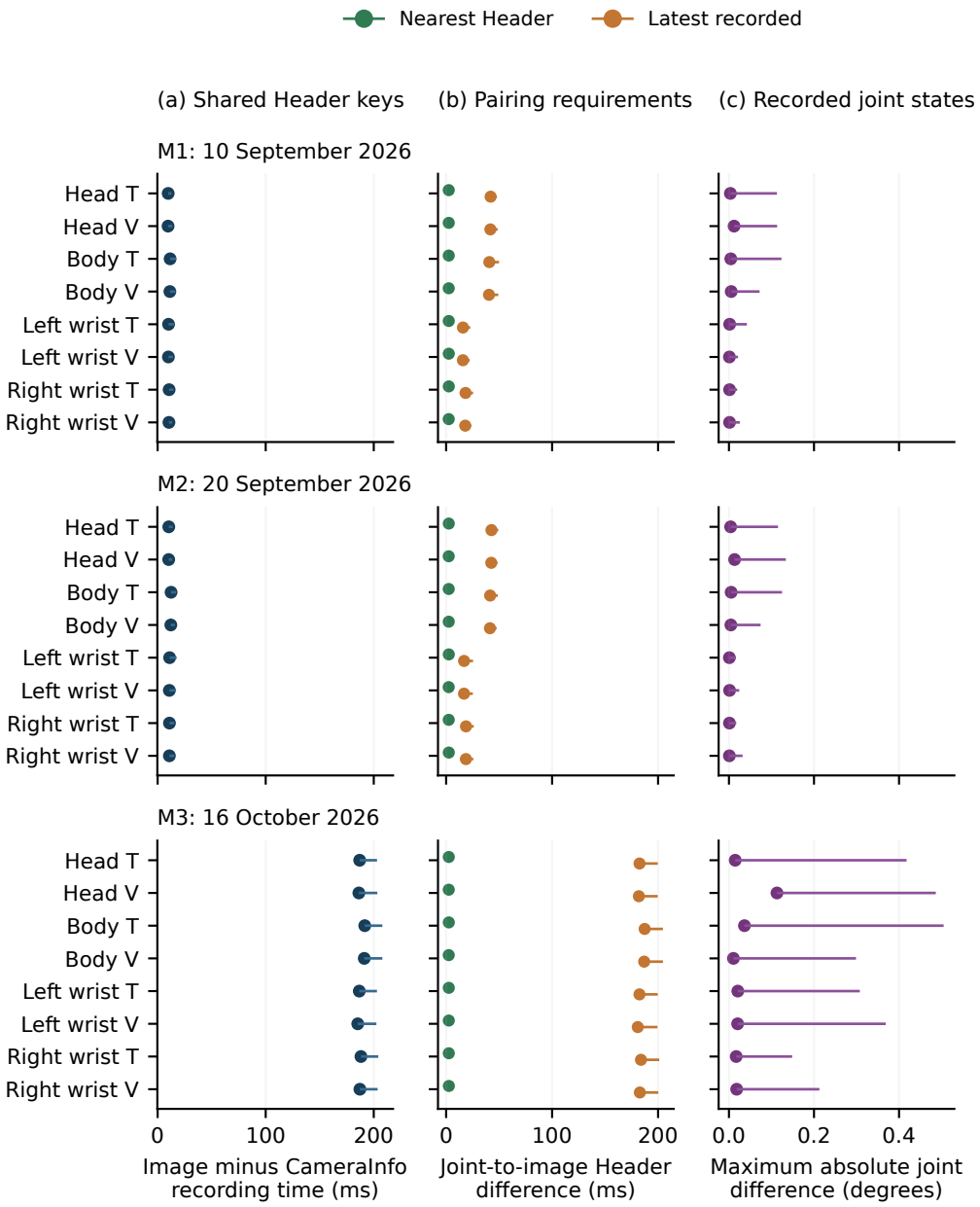


Figure 6. Session-stratified message and recorded-state audit. Rows identify M1–M3; T and V denote historical collection stages for head, body, and wrist roles. (a) Image-minus-CameraInfo recording time for exact shared Header keys. (b) Absolute nearest-header mismatch and signed latest-recorded joint-minus-image Header mismatch. (c) Maximum absolute recorded joint-position difference over 20 verified angular joints, for the fixed 1 Hz image subsample. Circles mark medians and lines extend to the 95th percentile (P95) in each distribution; they are not confidence intervals. Sessions, roles, stages, and messages are not independent robot replicates.

lookup targets the recorder's current joint information. Their timestamp differences and recorded configurations therefore need not coincide. The sampled state audit exposes this consequence in actual message values. Differences between recorded configurations remain distinct from errors against a physical reference. Larger October discrepancies co-occurred with greater temporal separation, but motion, host, and publisher histories were confounded. The archive therefore does not isolate a causal delay effect.

This distinction matters when assembling visual and proprioceptive observations. A model may require a state corresponding to an image, while a feedback process may require a current-state estimate. Neither a later joint Header nor a small nearest-header difference resolves that application requirement alone. In these records, later joint Headers were compatible with prior recording of the joint message. A controller's actual receive and use times remain separate measurements.

4.3. Spatial Interpretation and a Reusable Recording Contract

For a constant-speed, zero-order-held position scenario, age a implies an illustrative displacement va: 50 ms corresponds to 5 mm at 0.1 m/s or 25 mm at 0.5 m/s. An angular analogue is ωa. These relations explain why a timing budget depends on motion, but they do not transform the present logs into measured Cartesian or synchronization errors. The recorded joint differences add an empirical configuration-level comparison without supplying exposure timing, an independently measured robot pose, or a task outcome.

Temporal calibration methods relate physical measurements [7,8], and optical timing systems can observe exposure windows directly [3]. Host read timestamps and recorder metadata cannot replace those references. A regularly updated host log may still contain buffered or repeated images. Concurrent-stream unions likewise characterize the recorded all-stream age requirement, while the original cohort's absence of overlapping exceedance segments does not verify redundant sensors or exposure alignment.

A reusable sensing record should therefore preserve sensor identity and negotiated configuration, generation-related and host-read timestamps, a monotonic elapsed clock alongside wall time, device frame counters, and consumer use times when end-to-end age is the question. Content hashes or equivalent identity checks can expose repeated image content. This complements the acquisition provenance of synchronized visual–inertial datasets [9,10] and permits later readers to separate sensor, host, recorder, and consumer effects. The present byte manifests, declared timestamp layers, and explicit observation supports provide a narrower version of that contract for existing records.

4.4. Limitations and Discriminating Follow-Up

The main limitation is historical provenance. The roster-directed search broadened coverage beyond the initial five-computer selection, but recoverable files remain a convenience sample. SSH failures and inaccessible paths are not evidence that a robot lacked data. Distinct hashes, long durations, and large message counts do not establish independent cameras or fleet-wide reliability. The September sessions share a host; camera roles and collection stages are repeated information within each session. Conditional block-resampling ranges quantify within-record variation under the chosen blocks, not population uncertainty or future failure probabilities.

The stability logs lack verified historical camera identities, actual exposure and negotiated resolution, command lines, runtime versions, USB topology, and contemporaneous load. Current source agreement and reconstruction of stored fields support a calculation path without proving the historical invocation. A positive wall-clock adjustment can also resemble a long read interval. Durations and age integrals are conditional on that clock; the observed support ends at the last complete row. Pure trailing-NUL removal preserves

complete numeric prefixes but cannot recover unwritten observations or explain the suffixes. The retained-save comparison remains observational and does not isolate write duration or other logger effects.

MCAP metadata provides stronger message-level provenance, but neither Header nor publish time is a verified exposure clock. Recorder order does not establish controller receive order. Exact ranges, indexes, summary checksums, and independent deserialization validate retrieved metadata rather than unread image payloads or complete-bag hashes. The fixed 1 Hz state subsample cannot bound every image-time discrepancy, and the motions recorded during calibration do not represent arbitrary high-speed robot tasks. The shared-header cross-check and expanded-cohort analyses were revision-stage explorations, with specifications and amendments retained.

The most discriminating extension would combine a common exposure/reference clock with host and consumer traces, sensor identities, and configuration-matched repeated sessions. That would distinguish elapsed delivery gaps from clock artifacts and reveal whether regular updates carry current image content. An independent pose or task reference would then permit a spatial or performance claim. Those measurements were not acquired here.

5. Conclusions

The retrospective audit separates update recurrence, temporal requirements, and message availability. The original near-30 Hz cohort contained 326 intervals above 50 ms over 438.19 stream-hours, totaling 4.869 s of above-budget residence. The 234.92 stream-hour long-record extension added both near-30 Hz records with sparse events and a near-15 Hz record whose cadence alone implied approximately 25% residence above the same budget. Periodic-summary disagreement therefore depends on the timing regime and the requirement.

Across three message sessions, shared Header keys coexisted with session-dependent recording separations, and nearest-header versus latest-recorded joint pairing selected different declared times and recorded configurations. The complete selected Header audit and the fixed 1 Hz state comparison make those distinctions reproducible. They support preserving timestamp layers and choosing pairing rules according to the intended information requirement; exposure synchronization, Cartesian accuracy, and downstream robot-task performance remain outside the available measurements.

Supplementary Materials: Supplementary Information: short-record inventory; age and resampling diagnostics; image-save associations; image–joint timestamp diagnostics; the exploratory cadence extension; recorded joint-state discrepancies; provenance and reproduction details. Source data tables and offline analysis scripts are supplied in the accompanying reproducibility package.

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Data Availability Statement: The analysis scripts, parameter file, byte-integrity manifest, input digests, filename indexes, MCAP timestamp-range archives, and numerical result tables have been assembled in a local reproducibility package. The underlying records originated from organizational robot computers. Data ownership, redistribution permission, repository location, and the

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